

MarketMind AI - Milestone 2 Team Summary

A short handoff covering what we built, how the pieces connect and what every teammate should know.

DATASETS	ML + PIPELINE	DATABASE + API	ROLE DASHBOARDS
Clean sales, customer, inventory and demand sources	Features, model comparison, versioned outputs and safe imports	SQLAlchemy records served through secured FastAPI contracts	Owner, Manager, Executive and Administrator views

1. What Milestone 2 delivers

Customer intelligence RFM, purchasing behaviour, engagement, returns and three customer segments: Champions, Needs Attention and Hibernating.	Business forecasting Business Owner receives daily net INR revenue forecasts for 7, 14 and 30 days.	Operational forecasting Store Manager receives product demand and stock-risk context; Sales Executive receives a personal forecast.	Administrator monitoring Administrator sees model versions, algorithms, metrics, health and import jobs after MFA.
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2. Dataset and model map

Feature	Dataset	Selected model	Why / current value
Customer segmentation	Online Retail II	K-Means; Hierarchical comparison	No labels; 5,878 customers, 3 segments, Silhouette 0.364
Business revenue	Amazon sales	Weekday-aware Linear Trend	Lowest current MAE; 7/14/30-day INR forecast
Personal sales	Processed sales	Seasonal Naive	Small history favours recent weekly pattern; seller-scoped
Product demand	Train/eval Parquet	XGBoost Regressor	Non-linear context features; MAE 2.152, R2 0.918

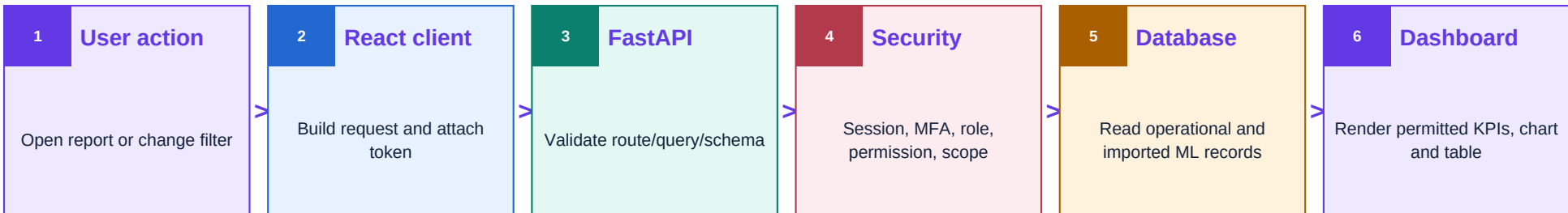
Simple project flow

Cleaned data -> feature engineering/model comparison -> versioned prediction and report -> duplicate-safe database import -> secured FastAPI endpoint -> role-specific React dashboard.

How the System is Connected

The frontend never reads the database directly. The backend is the security and integration boundary.

3. API and security workflow



API	Purpose	Allowed scope
POST /api/v1/auth/login	Issue access + refresh token	All roles; Administrator supplies MFA
GET /dashboard/access	Return allowed modules/actions	Current authenticated user
GET /customer-segments/summary	Segment and behaviour KPIs	Business, store summary or assigned scope
GET /forecasts/revenue	Business revenue series and metrics	Business Owner / Administrator
GET /forecasts/personal	Personal sales series	Signed-in Executive / Admin-selected seller
GET /forecasts/demand	Product demand, mapping and risk	Assigned Manager / Admin-selected store
GET /forecasts/monitoring	Models, health and jobs	Administrator after MFA

4. Role access

Role	Milestone 2 access	Restriction
Business Owner	Business segments + revenue forecast	No AI model configuration
Store Manager	Assigned-store summary + product demand	No business revenue or customer membership
Sales Executive	Assigned customers + own forecast	No other seller/store/business forecast
Administrator	All report types + monitoring after MFA	Store/seller selectors remain tenant-bound

Testing, Status and Team Demonstration

What is verified now, what should be stated honestly and how to demonstrate the work.

5. Current verification

Area	Result	What it covers
Backend	17 tests + Ruff passed	Auth/MFA, RBAC, scope, APIs, imports and Admin workflows
Preprocessing/ML	10 tests passed	Cleaning, features, clustering, candidates, horizons and artifacts
Database	Alembic check passed	Models and migrations remain aligned
Frontend	Oxlint + production build passed	Real API UI compiles; only non-blocking bundle-size warning

Important validations

Expired token refresh and retry; 403 for wrong role; 422 for invalid horizon/filter; 404 for missing model; duplicate-safe import; no false inventory mapping.

Demonstration order

1. Login as a role.
2. Open its segment or forecast report.
3. Change a filter/horizon.
4. Show the matching Swagger response.
5. Switch role and show a restriction.
6. Show Admin monitoring after MFA.

Current status

Milestone 2 is functionally integrated for local evaluation. Production deployment remains the final stage after HTTPS, secrets, backups, CI/CD, monitoring, confirmed data contracts and longer histories.

What every teammate should remember

MarketMind Milestone 2 is not only an ML notebook or a frontend mock-up. It is one connected workflow: data is cleaned, models are compared, results are versioned and imported, FastAPI applies backend security and React shows only the information allowed for the logged-in role. Revenue accuracy and demand-unit limitations must be explained honestly as data-hardening work.